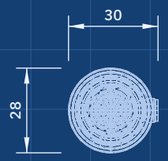
CONCEPT, NOT FOR FABRICATION

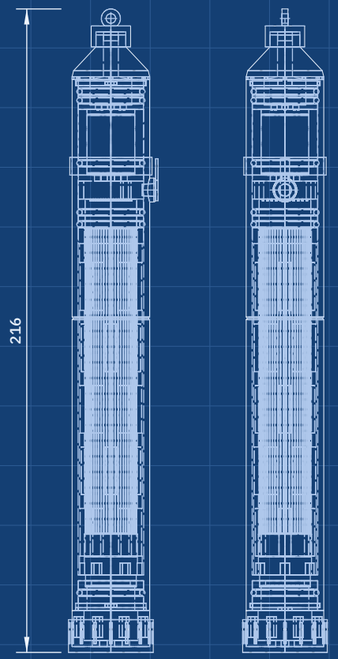
Helpful Water A: filter straw · github.com/BoujeeEnjinia1701/helpful-water

REV	DESCRIPTION	DATE	BY
P1	Concept sheet from the TRL 3 model	2026-11-11	AC
P2	Flows at 20 C after the independent STEM review	2026-11-11	AC



TOP VIEW

Scale 1:2; looking down (along -Z)

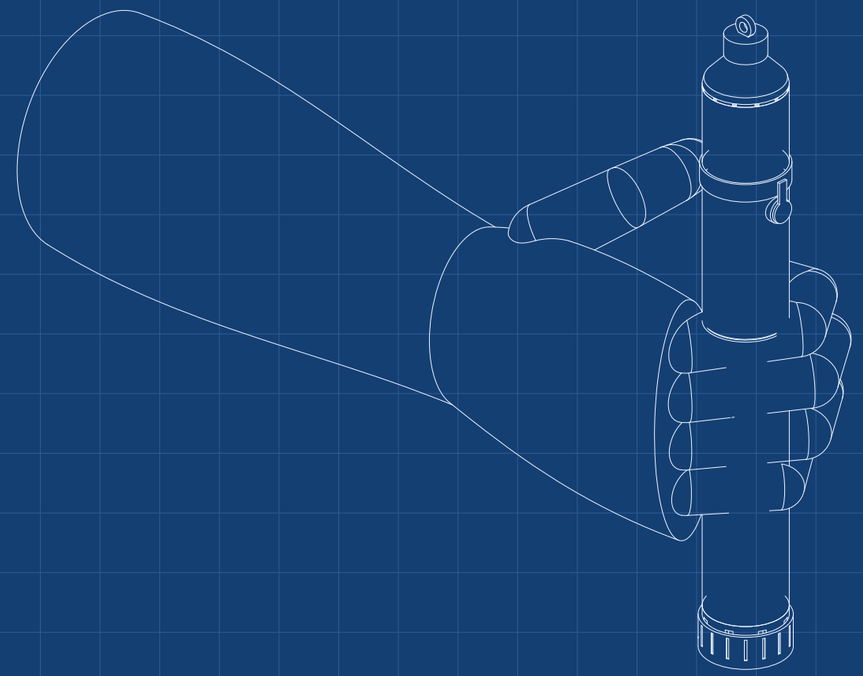


FRONT VIEW

Scale 1:2; looking at the front (along +Y)

RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; grey is an adult hand for scale; seen from the front right and above, about 30 deg elevation

KEY FIGURES (ESTIMATES, HWA-CAL-001)

205 mm long, 26 mm body, 29.6 mm across the port; 66 g
About 220 fibres 0.8/0.5 mm, 0.056 m2, outside-in
Flow at 10 kPa, clean: 110 (0.1 um) to 179 (0.2 um) mL/min
Child at 5 kPa: 45 to 74 mL/min
Carbon 16 x 20 mm, 1.9 g, contact 1.2 s at 200 mL/min
Backflush socket on the clean header; carbon bypassed
Prototype parts USD 17.46 (target USD 25)
Reduces bacteria and protozoa by design; no virus claim

HELPFUL WATER A: FILTER STRAW

github.com/BoujeeEnjinia1701/helpful-water

Hollow-fibre filter straw with backflush (option A1)

DRAWING NO.	REV	SHEET	SIZE
HWA-DWG-010	P2	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:2	mm		2026-11-11	Amish Chadha

MATERIAL / NOTES

Detailed TRL 3 model for concept communication

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